Introduction to Computational Chemistry & Biochemistry

Homework 2: Molecular Mechanics & Dynamics

Recommend using Jupyter (Colab) Notebook with RDKit and py3Dmol for basic molecular representations, molecular mechanics, and computational analysis and reporting. Recommendation for molecular dynamics is OpenMM or GROMACS (both available in Jupyter/Colab, ASU Sol Supercomputer and CalcFlow).

1. Most molecular building and viewing tools use basic molecular mechanics and dynamics to optimize molecular geometries when generating 3D molecular models. This is typically performed using either the Universal Force Field (UFF) or the Merck Molecular Force Field (MMFF94). One critical point to understand is that most molecular geometry optimizations use a gradient dissent method which finds a local minimized structure, which is definitely not guaranteed to be the global minimum energy structure. A clear and often used benchmark of multiple local versus global molecular structure geometries can be seen in the classic case of gauche and anti conformers of butane. Show that the starting initial molecular geometry leads to geometry optimization into either gauche and anti conformers of butane. Report the XYZ molecular representations for the starting geometry and optimized geometry along with the associated computed energy and all relevant geometry optimization computational information and parameters.

2. Do a torsional (dihedral) angle molecular mechanics computational scan of butane using two different force fields (e.g., UFF, MMFF94, AMBER, ...) as a comparison of which better reproduces the known energy barrier difference between the gauche and anti conformations of butane (0.75 +/- 0.15 kcal/mol). Report a well-formatted visual figure, showing the butane molecular configurations and torsional angle plot (relative energy in kcal/mol versus butane dihedral angle), with a figure caption providing all relevant computational information and parameters.

3. Extend the molecular mechanics/dynamics computational exploration of butane conformers by comparing 1-fluorobutane and 1-chlorobutane. Show how the fluoro and chloro substitution affects torsional dynamics and relative population of anti and gauche conformers at room temperature in the gas phase. Reporting should be a publication styled figure and associated caption (as detailed in the previous problem).

4. Alanine dipeptide (typically modeled as N-acetyl-L-alanine N'-methylamide, Ac-Ala-NHMe) serves as the primary benchmark model in computational chemistry and biochemistry to study protein backbone folding and torsional dynamics. Setup and run vacuum (gas-phase) alanine dipeptide molecular dynamics (MD) simulations. To check the quality, stability, and convergence of a molecular dynamics simulation, it's common to track/plot the time evolution of several key thermodynamic and structural parameters: Temperature, Density, Energy, Root Mean Square Deviation (RMSD), Root Mean Square Fluctuation (RMSF), and Radius of Gyration. In peptides/proteins it is also common to report Dihedral Angle Histograms and probability distributions (i.e., 2D Ramachandran style plot). The primary goal of this exercise is to show how a molecule explores its energy landscape by tracking its two primary backbone dihedral angles: φ (phi) and ψ (psi), i.e., the Ramachandran plot. However, it is important to also show all of the common time evolution plots mentioned above. Practically, there are a lot of parameters and conditions to consider when setting up MD runs, e.g., (i) Force Field Selection, (ii) Environmental Controls (Thermodynamics), (iii) Time Parameters (Integration Step & Duration), (iv) Treatment of Solvent (Implicit vs. Explicit vs. Vacuum), (v) Non-Bonded Interactions & Cutoffs, and (vi) Saving Frequency (Data Management). Students making reasonable and well-thought out choices is a major part of learning computational chemistry and biochemistry. This model system was chosen as a starting point, because it is computationally easy and fast to run, so students can vary parameters to get first hand experience with the practical effect on simulations and associated results.

5. To emphasize the importance of doing multiple simulations, starting with conditions that run computationally quickly, before moving to more complex, but often more realistic, simulations: Repeat molecular dynamics simulations on Alanine dipeptide but using an explicit water solvent environment. Again, the primary goal is to show how this dipeptide explores its energy landscape by tracking its two primary backbone dihedral angles: φ (phi) and ψ (psi), i.e., the Ramachandran plot. However, all of the common time evolution plots mentioned above should be reported, and students should become familiar with using these standard plots as a way to check the quality, stability, and convergence of a molecular dynamic simulation. Warning, a large explicit solvent environment will 10-1000x the computational time and size of the trajectory atomic coordinate files. Most modern MD packages will give you an estimate of simulation time and trajectory file size. Please look at these estimates to make practical decisions on time parameters, number of explicit solvent molecules and simulation 'box size', and saving frequency.